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DESIGN AND DEVELOPMENT OF SOLAR-POWERED UNMANNED AIR VEHICLE

A PROJECT REPORT

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

AEROSPACE ENGINEERING



Chandigarh University

Mohali, Punjab

November 2024



BONAFIDE CERTIFICATE

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ABSTRACT

5 A quadcopter is a sophisticated, incredibly agile aircraft with a straightforward construction and design. It has several qualities, including a great weight bearing ability. The next generation of helicopters, called quadcopters or quadrotors, have more dynamic stability than helicopters. One type of UAV (unmanned aerial vehicle) that can move objects quickly, cheaply, and without requiring much room or power is the quadcopter. It may be used for lifting, carrying, observing, rescuing, and data collection. It may be applied to military activities as well as surveillance. By figuring out and examining the dynamics of the quadcopter, the static and dynamic properties of its construction have been examined. Choosing the structure and materials that will satisfy the system's requirements for stiffness and strength. The objective is to maximize the performance characteristics and structural integrity of the frame while reducing weight and material consumption. The finite element method allows for efficient design iterations and improvements by providing a solid basis for simulating the behaviour of structures under various loading conditions.

5 This paper first explains the general architecture of the quadcopter frame model before using ANSYS to analyse the frame. And emphasizes the quadcopter's aerodynamic impact as well. It covers every facet of quadcopter operation, from electronics to mechanical design. Using a variety of formulas from research publications, it offers support for the choice of various components.

ABBREVIATION

UAV: Unmanned Aerial Vehicle

BLDC: Brushless Direct Current

P: Proportional

MATLAB: Matrix Laboratory

M-file: MATLAB text editor file

ESC: Electronic Speed Controller

GPS: Global Positioning System

PLA: Polylactic Acid

UAS: Unmanned aircraft system

RPA: Remotely piloted aircraft

BVLOS: Beyond visual line of sight

VTOL: Vertical take-off and landing

OSC: Organic solar cel

Cd: Drag coefficient

Table of Content

Acknowledgement.....	iii
Abstract.....	iv
Abbreviation.....	v
CHAPTER 1: INTRODUCTION	1
1.1 Aim of the project.....	1
1.2 Problem Statement.....	2
1.3 Identification of Issue	2
1.4 Identification of Problem	4
CHAPTER 2: LITERATURE REVIEW	6
CHAPTER 3: DESIGN CONCEPT	9
3.1 Design Ideas	9
3.2 Design Comparison	14
3.3 Selection of design	15
CHAPTER 4: NUMERICAL ANALYSIS AND CALCULATION	17
4.1 Weight Estimation	17
4.2 Calculating Thrust	19
4.3 Manual calculation of Thrust and Propeller diameter	19
4.4 Motor calculations	19
4.5 Calculating flight time based on Battery Capacity.....	20
4.5.1 Calculation is for Drone	21
CHAPTER 5: MATERIAL AND EQUIPMENT SELECTION	22
5.1 DC motor.....	22
5.2 Propellers.....	23
5.3 Electronic Speed Controllers (ESC's)	23
5.4 Lithium-Ion Polymer (Li Po) Battery.....	24
5.5 Channel Transmitter-Receiver.....	24
5.6 Flight controller.....	25
5.6.1 MPU-6050 (3-Axis Accelerometer and Gyro Sensor)	26
5.6.2 Arduino Uno R3	26
5.7 Solar cells	27
5.8 MPPT	27
5.9 Power distribution Board	28
5.10 Body Frame	28

CHAPTER 6: DETAIL DESIGN	29
6.1 Parts Design	29
6.1.1 Arms of the drone	29
6.1.2. Top Plate	30
6.1.3. Bottom Plate	30
6.1.4. Solar cell plate	31
6.2 Parts Assembly	32
6.2.1 Procedure	32
6.2.2 Final Design After Assembly	32
6.3 Structural Analysis	33
CHAPTER 7: DEVELOPMENT AND ASSEMBLY	37
7.1 Components Assembly	37
7.2 Circuit Description and Assembly	38
7.2.1 Circuit Connection and Its Description	38
7.2.2 Complete circuit diagram	39
CHAPTER 8: RESULT AND CONCLUSION	40
8.1 Result	40
8.2 Conclusion	43
8.3 Future Scope	44
8.4 Challenges	45
REFERENCES	46
APPENDIX	47
COST ANALYSIS	47
SPECIFICATIONS	48

Table of Figures

Figure 1.1: Quadcopter Drone.....	2
Figure 1.2: Overview of Drone	4
Figure 3.1: Delta Wing Drone	10
Figure 3.2: Glider	11
Figure 3.3: Quadcopter.....	12
Figure 5.1: . Brushless DC motors (BLDC)(1000kv)	22
Figure 5.2: Propellers	23
Figure 5.3: ESC	24
Figure 5.4: Li-Po Battery	24
Figure 5.5: RC and receiver	25
Figure 5.6: MPU 6050.....	26
Figure 5.7: Arduino UNO.....	26
Figure 5.8: Solar cell	27
Figure 5.9: MPPT.....	27
Figure 5.10: PDB.....	28
Figure 5.11: PLA filament.....	28
Figure 6.1: Arm	30
Figure 6.2: Base Top	30
Figure 6.3: Base Bottom.....	31
Figure 6.4: Solar Cell Plate	31
Figure 6.5: Final Assembly	32
Figure 6.6: Assembly of Arm and Plate only	33
Figure 6.7: Equivalent Elastic Strain.....	34
Figure 6.8: Maximum Principal Elastic Strain.....	34
Figure 6.9: Maximum Principal Stress.....	35
Figure 6.10: Strain Energy	35
Figure 6.11: Total Deformation	36
Figure 7.1: Complete Circuit Diagram.....	39
Figure 8.1: Phase-1 of Assembling 3-D printed body frame.....	40
Figure 8.2: Implemented Solar Panels over the top view of the UAV body frame.....	41
Figure 8.3: Assembly of electrical components inculcated with wire connections	41
Figure 8.4: Attached Propellers along with solar panels, DC Motors.....	42
Figure 8.5: Top View of Solar Powered UAV	42

Table of Standards

Table 1: Design Comparison	14
Table 2: Weight Comparison	17
Table 3: Cost Analysis.....	47

CHAPTER 1

INTRODUCTION

For the past few decades, mankind has grown a lot in aerospace industry, now sky is no longer the limit. Manufactures are constantly trying to get a more effective flight and ways to make the use of the finite resources infinitely as possible. The integration of solar energy into unmanned aerial vehicles (UAVs) marks a significant step forward in drone technology, with the potential for longer flight periods and less reliance on traditional fuel sources. This study focuses on the design and experimental testing of a solar-powered drone, investigating its capabilities, efficiency, and operational feasibility. The drone intends to enhance durability while minimizing environmental impact by capturing solar energy via photovoltaic cells, making it suitable for a wide range of uses, including surveillance and environmental monitoring.

This research looks into the engineering fundamentals underlying solar drone design, such as material selection, power management systems, and aerodynamic factors. It also investigates the difficulties connected with energy storage and conversion, notably during periods of low sunshine or bad weather.

1.1 Aim of the project

This study aims to solve practical issues like weight optimization and structural integrity while demonstrating through experimental validation how solar energy can improve drone performance.

In the conclusion, our research clears the path for next advancements in sustainable UAV technology in addition to adding to the expanding field of renewable energy applications in aviation. Researchers and engineers hoping to push the limits of what is feasible with solar-powered aircraft systems will find great value in the findings. Solar cell implementation on UAVs is a widely used technique among academics, researchers, and hobbyists. UAVs employ solar cells attached to a power electronic circuit to power the engine and charge the battery, which serves as a buffer during nighttime or cloudy conditions. Advancements in remote sensing technology, as well as private and military funding, have made UAVs more competent and cost-effective.



Figure 1.1: Quadcopter Drone [12]

1.2 Problem Statement

Nowadays, the ability to fly without using conventional fossil fuels has received a lot of attention in recent years, both in terms of application and science, because the main concerns are rising global temperatures and dwindling natural resources. Since then, the use of electric aircraft has grown, but the key issue is their high-power consumption in comparison to their limited energy storage capacity, resulting in an endurance that rarely exceeds half an hour. Increasing the size of the battery or incorporating more batteries increases the plane's weight, which has an inverse relationship with the UAV's flight time.

Thus, a major concern in both UAVs and commercial aviation has been the ability to fly an aircraft to raise the endurance limit.

Using solar cells to harness infinite sun energy is one way to extend the flight time. By connecting these cells to an electronic circuit, they can supply enough power for the motor and other electronics. If there is extra, it can also be stored in the battery, which can then be used as a buffer when flying in cloudy or dark conditions. Therefore, employing solar-powered aircraft with electric-based propulsion systems, which can do away with fuel, is one potential way to increase endurance. These aircraft receive electricity continually throughout the day from the limitless solar energy and also solve the limited energy storage capability problem.

1.3 Identification of Issue

Designing solar-powered drones presents several key challenges that must be addressed to enhance their performance and viability. These challenges include:

- **Energy Collection Efficiency:** Solar panels must transform sunlight into useful electricity with high efficiency. Current solar technologies achieve only about 21% efficiency, requiring greater surface areas to capture enough energy for sustained flight. This need may increase drone size and weight, complicating design concerns.
- **Weight Management:** The drone gains a substantial amount of weight due to the incorporation of solar panels and energy storage technologies. For flying stability and efficiency, a good power-to-weight ratio must be achieved. In order to ensure that the drone can carry out its intended cargo without sacrificing performance, engineers must balance the weight of solar cells, batteries, and other components.

Drones that run on solar power have solar panels built onto their body or wings. The drone's propulsion engine and onboard electronics are powered by the electrical energy produced by these panels, which absorb sunlight. Since it necessitates effective energy conversion and a balance between power generation and drone weight, integrating solar panels into drones is a challenging engineering task.
- **Weather Dependency:** Solar-powered drones rely largely on weather conditions. Overcast skies, rain, or nighttime operations can significantly limit their flight capabilities since they may not generate enough energy to continue operations. This dependency is a barrier for applications that require consistent performance independent of external conditions.
- **Space Constraints:** Many drones have limited space for installing solar panels and backup batteries. This limitation affects both the design of the drone and its payload capacity, making it difficult to optimize for various operational needs while maintaining sufficient energy supply.
- **Structural Integrity:** The design must ensure that the drone is lightweight yet robust enough to withstand atmospheric turbulence and other stresses during flight. This involves selecting appropriate materials and engineering solutions that maintain durability without adding excessive weight.
- **Flight Stability and Control:** Achieving stable flight with a solar-powered drone involves complex engineering challenges, including the precise alignment of wings and control surfaces to maximize solar energy absorption while maintaining aerodynamic efficiency. Research is ongoing to optimize these designs for better performance.

- **Advanced Energy Management:** Efficiently managing the energy generated by solar panels, including storage and distribution systems, is critical for maximizing flight time and operational effectiveness. Developing advanced energy management systems that can adapt to varying sunlight conditions is an ongoing area of research.

While addressing or having solutions of these challenges is essential for the successful development of solar-powered drones that can operate effectively across various applications, from environmental monitoring to disaster response, while contributing to sustainability goals through reduced carbon emissions.

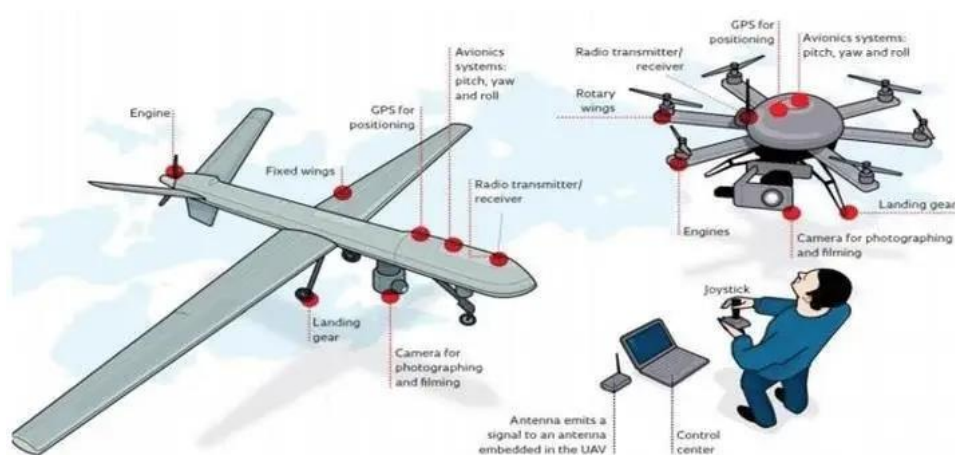


Figure 1.2: Overview of Drone

1.4 Identification of Problem

The development of solar-powered drones presents various challenges that need to be addressed to enhance their efficiency, reliability, and applicability. Followed below are the key problem areas identified in the design and experimental analysis of these unmanned aerial vehicles (UAVs).

- **Structural Integrity and Design Challenges:** Solar UAV structural design must consider both aerodynamic efficiency and solar panel integration while maintaining performance. The frame must be light but strong enough to withstand external forces during flight. Furthermore, the positioning of solar cells on the drone's wings requires careful attention to maximize exposure to sunlight while maintaining aerodynamic integrity.

- **Integration of Advanced Technologies:** The use of advanced technologies such as Maximum Power Point Tracking (MPPT) systems is critical for optimizing energy consumption from solar panels. However, implementing these systems adds complexity to the design and control mechanisms, which can lead to failures if not managed properly. Ensuring seamless integration of power management and flight control systems is critical for improving overall drone performance.
- **Flight Time Optimization:** The quest for the longest possible flight time remains a critical issue. Solar drones must balance the weight of their components against the amount of energy they can harvest. The design must ensure that the energy collected during daylight hours exceeds the amount consumed during flight. This requires complex calculations for weight distribution, lift generation, and energy consumption rates. Drones may fail to achieve the desired endurance levels for various applications if effective optimization strategies are not implemented.
- **Regulatory and Safety Concerns:** As with any UAV technology, regulatory frameworks governing airspace usage pose challenges for the deployment of solar-powered drones. Compliance with safety standards while ensuring efficient operation in various applications—such as surveillance, delivery services, or environmental monitoring—requires ongoing research and adaptation to evolving regulations.

CHAPTER 2

LITERATURE REVIEW

Santhosh Chandra and Kirankumar (2023) (Developing a Quadcopter Frame and its Structural Analysis). [1] A quadcopter is one of the most versatile unmanned aerial vehicles (UAVs) for a variety of uses. Pitch, roll, Motion, yaw, and the fundamental technique are all controlled by of operation in quadcopters with symmetry. This undertaking makes use of a DJI F450 quadcopter that has been tuned frame layout. Fusion 360 allows each element of the frame was created. The recommended design is validated for viability utilizing the results of Fusion 360's stress study, and an appropriate material for production is selected by comparing it to five different types of materials.[1]

Çaşka, et al (2020) (Design and analysis of quadcopter drone frame using finite element analysis). [2] The finite element method was employed in this study to create a small unmanned aerial vehicle (UAV) that could be equipped with landing platforms and to analyze the design statically. When stationary analysis, the foundation is steady, and the mass of the load is considered the quadcopter body. Acrylonitrile ABS (butadiene styrene) and PLA (polylactic acid) Materials that are 0.05 mm thick are used to make every part of the quadcopter that is utilized in the i3 Prusa for Finite Element Analysis (FEA) printer. Additionally, the ABS quadcopter frame's distortion was measured at 0,014 mm, whereas the PLA frame had the same distortion revealed a 0,010mm deformation.[2]

Sarika Marneni, Chinavath Soniya, H Krupa, (2024) (Design and analysis of quadcopter drone frame using finite element analysis) [4] AUTODESK FUSION 360 CAD model of the quadcopter drone frame was created. The design of the product takes into consideration criteria provided by the Directorate and drone DGCA, or General of Civil Aviation. The rotor base, i.e. the distance measured between two rotors, is regarded as the drone's length measures 250 mm. The planned quadcopter drone can be used for racing or exploration based on the requirements of the electronic component. Next, the intended frame is simulated to check for structural as well as modal behavior, for which the outcomes are calculated under many scenarios.[4]

Prof. A.V. Javir, Ketan Pawar, Santosh Dhudum, Nitin Patale, Sushant Patil (2015) (Design Analysis and Fabrication of Quadcopter)[6] The final assembly of the quadcopter is produced in a CAD model by carefully examining the design of the machine and the materials used for each part. The quadcopter assembly is analyzed in three different ways. The first structural analysis's findings indicate that the model's deformation ranges from 0 m at the minimum to 1.317e006 m at the maximum. It's okay that there is distortion. Von Mises stress result is at least 1.9369 Pa, and the model is safe because it is operating within the limit. The second step is modal analysis, which is necessary to determine the inherent frequencies of various modes.[6]

Smith Jason (2017) (An Analysis of Solar Powered Unmanned Aerial Vehicle)[3] This Paper investigates the possibilities of solar-powered drones (UAVs) in a variety of applications, highlighting their capacity to increase endurance and minimize dependency on conventional energy sources. The study emphasizes the limitations of typical UAVs, particularly their reliance on batteries and fuel, which reduces flying time.[3]

Shah-Krutikk Nilesh-Kumar, Prajapati-Prince, Karandikar-Jay Mukul Bhai (2023) (Design and Development of Solar Drone) [5] The paper explores the integration of solar energy into unmanned aerial vehicles (UAVs) to enhance their operational efficiency and flight endurance. Conducted by a team from the Indus Institute of Technology and Engineering, the study focuses on a quadcopter that utilizes both solar panels and lithium polymer batteries to maximize flight output.[5]

Yuhei Chu, Chun Leung Ho, Yoonjo Lee, Boyang Li (2023) (Development of a Solar-Powered Unmanned Aerial Vehicle for Extended Flight Endurance)[9] The article explores the design and implementation of a solar-powered UAV aimed at enhancing flight endurance. Conducted as part of a capstone project at the Hong Kong Polytechnic University, the research focuses on modifying a commercially available remote-controlled UAV to integrate solar panels with battery storage. The primary goal is to significantly extend the UAV's flight time by utilizing solar energy. [9]

27 **Parvathy RAJENDRAN, Howard SMITH (2020) (Experimental Analysis of Small Solar Unmanned Aerial Vehicle to Predict Aerodynamic Performance)**[8] This paper presents an experimental analysis of a small solar-powered unmanned aerial vehicle (UAV) to predict its aerodynamic performance. The researchers developed a mathematical model for solar UAVs and fabricated a prototype model. They conducted bench and flight tests to compare the aerodynamic characteristics and performance of the solar and non-solar UAV prototypes. Both accelerated and un-accelerated flight test methods were used to predict the drag polar curve, and the results were compared with the simulation data from the mathematical design model.[8]

CHAPTER 3

DESIGN CONCEPT

The integration of delta wing, glider, and quadcopter designs in solar-powered unmanned aerial vehicles (UAVs) represents a significant advancement in aerial technology, aimed at enhancing flight efficiency and sustainability. Delta wings offer superior aerodynamic performance and stability, making them ideal for long-distance flights, while gliders capitalize on their lightweight structure to maximize lift and minimize energy consumption. On the other hand, quadcopters provide exceptional manoeuvrability and versatility, enabling complex flight patterns and operations in varied environments. By harnessing solar energy through advanced photovoltaic systems, these UAVs can achieve prolonged flight durations without the constraints of traditional fuel sources. This design concept not only addresses the pressing need for environmentally friendly aviation solutions but also optimizes the interplay of aerodynamics, energy management, and structural integrity, paving the way for innovative applications in surveillance, environmental monitoring, and disaster response.

3.1 Design Ideas

The design concepts of delta wings, gliders, and quadcopters represent innovative approaches in the realm of Unmanned Aerial Vehicles (UAVs), each offering unique advantages tailored to specific applications. Delta wings, characterized by their triangular shape, provide excellent aerodynamic efficiency, high lift at various angles of attack, and structural strength, making them ideal for high-speed missions and surveillance tasks. This design is particularly advantageous for UAVs that require stability during aerobatic manoeuvres and can accommodate larger fuel volumes for extended flight durations. Gliders, on the other hand, emphasize energy efficiency and long-range capabilities. Their elongated wings enable them to soar using minimal power by harnessing thermals and updrafts, making them suitable for applications such as environmental monitoring and aerial photography. Lastly, quadcopter designs integrate multiple rotors to achieve vertical take-off and landing (VTOL) capabilities, providing exceptional manoeuvrability in confined spaces. This versatility allows quadcopters to excel in tasks ranging from search and rescue operations to package delivery. Together, these UAV designs illustrate the diverse engineering strategies employed to meet the evolving demands of aerial technology.

- 24
1. **Delta wings** are a distinctive type of aircraft wing characterized by their triangular shape, resembling the Greek letter delta (Δ). This design offers several aerodynamic advantages, particularly for high-speed flight. Followed is an overview of the key aspects of delta wings:

Design Characteristics

- **Shape:** Delta wings are typically triangular and sweep back sharply from the fuselage, often with a straight trailing edge. This configuration allows for a larger surface area, which contributes to increased lift and reduced wing loading compared to conventional wings.
- **Angle of Sweep:** The leading edge of delta wings usually has a sweep angle greater than 60 degrees, enhancing their performance at high speeds and angles of attack.
- **Versatility in Operations:** Delta wings can support various operational modes, including **Vertical Take-off and Landing (VTOL)** and **Short Take-off and Landing (STOL)**. This versatility allows UAVs to be deployed in diverse environments, from urban settings to open fields.



Figure 3.1: Delta Wing Drone [12]

The delta wing design offers significant advantages for UAVs, combining aerodynamic efficiency with structural strength and operational versatility. These features make delta-wing UAVs a compelling choice for both military and civilian applications.

Additionally, delta wings exhibit a high stall angle, meaning they can sustain stable flight at greater angles before stalling compared to conventional wings. This characteristic makes them particularly resilient during aggressive manoeuvres, which is crucial for military UAV applications where agility

and control are paramount. Furthermore, the inherent structural efficiency of delta wings allows for a lighter design that can withstand aerodynamic stresses, contributing to overall stability during flight. As a result, delta-wing UAVs can achieve improved manoeuvrability and operational versatility, making them effective in various missions ranging from surveillance to combat operations.

2. **Gliders** represent a specialized category of Unmanned Aerial Vehicles (UAVs) designed to maximize aerodynamic efficiency and extend flight duration through passive lift generation. Unlike powered aircraft, gliders rely on their wing design and air currents to maintain altitude, making them particularly suited for applications requiring long-range surveillance, environmental monitoring, and scientific research. The aerodynamic structure of gliders typically features long, slender wings that enhance lift-to-drag ratios, allowing them to soar for extended periods without the need for continuous propulsion. This efficiency is further complemented by their ability to exploit thermals and updrafts, enabling them to gain altitude without expending energy. Recent advancements in UAV technology have led to the development of sophisticated glider designs that integrate autonomous navigation systems and advanced payload capabilities, facilitating operations in challenging environments such as the stratosphere. As a result, gliders are increasingly utilized in various sectors, including meteorology, wildlife tracking, and remote sensing, where prolonged flight and minimal operational costs are paramount.



Figure 3.2: Glider [12]

3. **Quadcopters** have revolutionized the field of Unmanned Aerial Vehicles (UAVs) by offering a versatile and efficient platform for various applications, from aerial photography to surveillance and delivery services. Characterized by their four rotors, quadcopters provide exceptional stability and manoeuvrability, allowing them to perform complex flight patterns and hover in place with ease. This design enables them to operate effectively in confined spaces and challenging environments, making them ideal for both commercial and military operations. The quadcopter's ability to achieve vertical take-off and landing (VTOL) eliminates the need for runways, enhancing their operational flexibility. Additionally, advancements in technology have led to the integration of sophisticated navigation systems, sensors, and payload capabilities, further expanding their utility. As a result, quadcopters are increasingly utilized across diverse sectors, including agriculture for crop monitoring, emergency response for search and rescue missions, and infrastructure inspection for maintenance assessments. Their combination of agility, ease of use, and adaptability positions quadcopters as a cornerstone of modern UAV technology.



Figure 3.3: Quadcopter [12]

The key components of a quadcopter's frame are essential for ensuring structural integrity, stability, and functionality during flight. These components include:

- **Frame Structure:** The frame serves as the primary skeleton of the quadcopter, providing support for all other components. It typically consists of lightweight materials such as carbon fibre, aluminium, or fiberglass to minimize weight while maintaining strength. The

design can vary, with common configurations including X-frame and H-frame designs that optimize aerodynamics and load distribution.

- **Arms:** Each quadcopter has four arms extending from the central frame, each housing a motor at its end. These arms must be rigid enough to withstand the forces during flight while being lightweight to enhance overall performance. The length and angle of the arms are critical for ensuring proper motor placement and effective thrust generation.
- **Electronic Speed Controllers (ESC):** While not part of the frame itself, ESCs are mounted on or near the frame to control motor speed based on commands from the flight controller. Their placement is crucial for efficient wiring and minimizing weight distribution issues.
- **Improved Efficiency:** The combination of solar energy with battery storage can lead to a significant reduction in battery discharge rates. Studies have shown that solar-powered quadcopters can save **an average of 22.5% of battery capacity** during flight, which translates to longer operational times. This efficiency is especially beneficial for applications requiring prolonged surveillance or monitoring.
- **Extended Operational Range:** With solar panels providing additional energy, quadcopters can achieve longer ranges without needing to return to base for recharging. This capability opens new possibilities for remote operations in areas where access to charging stations may be limited.
- **Shared Power Load:** By incorporating solar panels, the quadcopter can share the power load between the solar cells and the batteries. This means that while the motors draw power for propulsion, some of that energy is replenished by the solar panels, leading to an overall increase in available power and extended flight duration.

As **the demand for sustainable and efficient aerial solutions** grows, solar-powered quadcopters **are poised to play a significant role** in various sectors, offering a cleaner alternative to traditional UAVs while enhancing operational capabilities and reducing reliance on fossil fuels.

3.2 Design Comparison

Table 1: Design Comparison

Design Aspect	Delta Wing UAV	Glider UAV	Quadcopter UAV
Aerodynamics	High lift-to-drag ratio at high speeds; efficient for long-range flights.	Excellent lift generation through soaring; optimized for energy efficiency.	Good manoeuvrability with vertical take-off and landing (VTOL) capabilities.
Flight Time	Extended flight time with solar integration; efficient at cruising speeds.	Exceptionally long flight times due to energy harvesting from thermals; solar panels enhance endurance.	Improved flight time through solar panels that recharge batteries during flight.
Manoeuvrability	Limited low-speed manoeuvrability; best suited for high-speed operations.	Limited agility; primarily designed for sustained gliding rather than rapid manoeuvres.	Highly agile and capable of complex manoeuvres, suitable for diverse missions.
Operational Flexibility	Requires longer runways; less flexible in confined spaces.	Best suited for open areas; limited in urban settings or confined environments.	Extremely versatile; can operate in various environments, including urban areas.
Applications	Ideal for military surveillance and long-range missions.	Suitable for environmental monitoring and scientific research over large areas.	Used in search and rescue, infrastructure inspection, and aerial photography.
Energy Efficiency	Efficient at high speeds but may struggle at lower speeds without adequate solar input.	Highly energy-efficient during gliding; solar panels significantly extend operational range.	Energy-efficient with solar panels, but battery life can be a limiting factor without sufficient sunlight.

The comparison of delta wing, glider, and quadcopter designs for solar-powered UAVs reveals distinct strengths and weaknesses tailored to specific operational needs. Delta wing UAVs excel in high-speed, long-range missions due to their aerodynamic efficiency but are less manoeuvrable and require longer take-off distances, making them suitable for military applications. In contrast, gliders are optimized for energy efficiency and prolonged flight times, leveraging thermal currents and solar energy, ideal for environmental monitoring tasks where endurance is crucial. Quadcopters, with their remarkable agility and VTOL capabilities, offer versatility in various applications, from search and rescue to infrastructure inspection, while benefiting from solar integration to enhance flight duration. The choice of design depends on the mission requirements: delta wings are preferred for speed and range, gliders for endurance and efficiency, and quadcopters for flexibility and manoeuvrability in diverse environments. The incorporation of solar technology across all three designs enhances their operational capabilities by extending flight times and reducing reliance on traditional power sources.

3.3 Selection of design

Quadcopters are increasingly recognized as efficient platforms for solar-powered UAVs due to several key factors that enhance their performance and operational capabilities.

1. **Energy Management:** The integration of solar panels allows quadcopters to harness solar energy during flight, which can be used to recharge batteries in real-time. This capability significantly extends flight duration by sharing the power load between the solar cells and the battery, reducing reliance on battery power alone. Previously have shown that solar-powered quadcopters can save an average of 22.5% of battery capacity during flight, leading to longer operational times.
2. **Lightweight Design:** Quadcopter frames are typically constructed from lightweight materials, which is crucial for maximizing the efficiency of solar energy utilization. A lighter frame reduces the overall power required for flight, allowing more energy to be drawn from the solar panels, thereby enhancing endurance.
3. **Agility in Flight:** Quadcopter designs provide excellent manoeuvrability, enabling them to execute complex flight patterns and hover in place. This agility is particularly beneficial for tasks that require precise positioning, such as aerial photography or search and rescue operations.

4. **Reduced Operational Costs:** By utilizing solar energy, quadcopters can reduce operational costs associated with battery replacements and charging infrastructure. This cost-effectiveness becomes increasingly significant over time as solar technology continues to improve, making solar-powered quadcopters a sustainable option for long-term use.
5. **Hybrid Power Systems:** The combination of solar energy and lithium polymer (LiPo) batteries allows quadcopters to maintain a steady power supply during both sunny and cloudy conditions. This hybrid approach enables them **to store excess energy generated during peak sunlight hours for use when** solar input is low, effectively extending their operational range without frequent landings for recharging.

In conclusion, the efficiency of quadcopters as solar-powered UAVs is enhanced by their ability to harvest renewable energy, advanced power management systems, lightweight construction, hybrid power capabilities, and reduced environmental impact. These factors collectively contribute to significantly improved endurance and operational range, making quadcopters an attractive option for various applications in fields such as environmental monitoring, agriculture, and disaster response.

CHAPTER 4

NUMERICAL ANALYSIS AND CALCULATION

Numerical analysis and calculation play a crucial role in the design and optimization of solar-powered unmanned aerial vehicles (UAVs), enabling engineers to develop efficient and effective models that maximize performance while minimizing energy consumption. By employing numerical methods, researchers can simulate various flight conditions, assess aerodynamic characteristics, and evaluate the impact of solar panel integration on overall vehicle stability and efficiency. These analyses facilitate the determination of key parameters such as power-to-weight ratios, thrust requirements, and energy storage needs, which are essential for achieving extended flight endurance. Furthermore, advanced computational techniques, including Maximum Power Point Tracking (MPPT) algorithms, allow for real-time adjustments to optimize energy harvesting from solar cells under varying environmental conditions. As the demand for sustainable aviation solutions grows, the application of numerical analysis in solar-powered UAV development becomes increasingly vital for enhancing operational capabilities and ensuring reliable performance in diverse applications.

4.1 Weight Estimation

Table 2: Weight Comparison

S.No.	Components	Quantity	Weight (gm)
1.	Motor (1000kv)	4	288
2.	Propeller	4	80
3.	ESC (30A)	4	92
4.	Solar cells	5	50
5.	Battery (Li-Po)	1	175
6.	Receiver	1	15

7.	Arms	4	280
8.	Base plate	2	100
9.	Solar Pannel	1	40
10.	Flight controller	1	20
11.	Landing Gears	1	60
12	Miscellaneous		50
	TOTAL WEIGHT		1250

Empty Weight,

$$\begin{aligned}
 W_e &= W_{st} + W_{pp} + W_{sys} \\
 &= 380 + 288 + 407 \\
 &= 1075 \text{ gm} / 1.075 \text{ Kg}
 \end{aligned}$$

Take-off gross weight,

$$\begin{aligned}
 W_o &= W_e + W_{bat} + W_{pay} \\
 &= 1075 + 175 + 250 \\
 &= 1500 \text{ gm} / 1.5 \text{ Kg}
 \end{aligned}$$

Empty Weight fraction,

$$\begin{aligned}
 W_e / W_o &= 1.075 / 1.5 \\
 &= 0.716
 \end{aligned}$$

4.2 Calculating Thrust

Thrust is the force required to drive or supply the motion to the propeller of Quadcopter. It is generated by the rotors which spin at certain angular velocity. Thrust developed by a rotor is,

$$T = \rho A V^2 \quad (1)$$

where,

ρ = real time air density of air (Kg/m³)

A = Cross sectional area of the propellers

V = peripheral velocity of rotors

The peripheral speed is given by the product of the circumference ($2\pi R$) and the speed (RPM).

4.3 Manual calculation of Thrust and Propeller diameter

Drone gross weight: 1.5kg and its thrust should be the 3 times the drone gross weight that is 4.5kg or 44.145N from this we find the diameter of the propeller i.e.

$$T = 44.145\text{N} \quad A = 3.14R^2$$

$$\rho = 1.225 \text{ kg/m}^3 \quad V = 31.4 \text{ m/s}$$

$$T = \rho A V^2$$

$$R = 10.77\text{cm} \sim 4.2 \text{ inch} \sim 4\text{inch}$$

Therefore, diameter of propeller is taken as ~8 inch

$$\text{Total Thrust} = 3 * \text{Total weight of Quadcopter} = 44.145\text{N}$$

Total number of motors: 4

$$\text{Total thrust by each motor} = \text{Total thrust} / 4 = 11.036\text{N}$$

4.4 Motor calculations

The motors should be selected in such a way that it follows following thrust to weight relationship.

$$\text{Ratio} = \text{Thrust} / \text{weight} = m a / m g = a / g$$

Thus, vertical takeoff and vertical landing (VTOL) is possible only when, $(a/g) > 1$ or in other words, The total thrust to total weight ratio should be greater than 1 so that the quadcopter can accelerate in the upward direction. In this case, we assumed that,

$$\text{Total Thrust} = 3 * (\text{Total weight of Quadcopter}) = 44.145\text{N}$$

$$\text{Total thrust by each motor} = \text{Total thrust} / 4 = 11.036\text{N}$$

4.5 Calculating flight time based on Battery Capacity

Maximum Flight Time (min.) = Battery Storage Amp Rating (mAh) x 60 Current Draw of UAV (Amps)
Recommended Flight Time (min.) = Maximum Flight Time (min.) x Safety Factor

Where: Safety Factor = 0.4 to 0.8 for Airplane type UAV's

0.4 to 0.7 for Multirotor type UAV's

Example 1 – Airplane type UAV: Assume an airplane-type UAV uses a 4S LiPo battery with a stamped rating of 5,000 mAh, and is observed using 4 amps on average when flying camera work. What is the maximum and actual flight times for this vehicle?

Step 1: The rating of the battery is 5,000 mAh

Step 2: The amp-minute for the battery is $5,000 / 1,000 * 60 = 300$ -amp minutes.

Step 3: From observation of the ground control station, the UAV uses 4 amps on average during flight.

Step 4: The maximum flight time for the UAV is $300 / 4 = 75$ minutes.

Step 5: The recommended flight time is 30 minutes (40 percent safety factor - 0.4×75 minutes) to 60 minutes (80 percent safety factor - 0.8×75 minutes).

Example 2 – Multirotor type UAV: A multirotor type UAV uses a 4S LiPo battery with a stamped rating of 11,000 mAh is uses 25 amps on average when flying with a full sensor load. What is the maximum and actual flight times for this vehicle?

Step 1: The rating of the battery is 11,000 mAh

Step 2: The amp-minute for the battery is $11,000 / 1,000 * 60 = 660$ -amp minutes.

Step 3: From observation of the ground control station, the UAV uses 25 amps on average during flight.

Step 4: The maximum flight time for the UAV is $660 / 25 = 26.4$ minutes.

Step 5: The recommended flight time is 10.5 (40 percent safety factor – 0.4×26.4 minutes) to 18.5 minutes (70 percent safety factor – 0.7×26.4 minutes).

4.5.1 Calculation is for Drone

Battery: 2200 mAh, 24.1V, 3cell 30C

Amp-minute = $2200/1000 \times 60$

= 132-amp minutes

UAV uses 20 amps on average during the flight

Maxi flight time = $132/20$

= 6.6 minutes

Recommended flight time = $0.4 \times 6.6 = 2.64$ minutes

= $0.7 \times 6.6 = 4.62$ minutes

But our drone will give more flight time as the charge of battery decreases the solar cells will recharge the battery and will continue its flight in air itself. 5 solar cells give almost 20V and it will provide more energy to our drone and increase the flight time.

Some of specification related to components like:

Solar cell: output of 20V

Motor: 12A/60S

ESC: 30A

Battery: 2200mAh, 24.1V

RC range: 1500m

CHAPTER 5

MATERIAL AND EQUIPMENT SELECTION

The selection of materials and equipment is a critical aspect in the design and development of solar-powered unmanned aerial vehicles (UAVs), as it directly influences their performance, efficiency, and operational capabilities. This process involves evaluating various components, including photovoltaic cells, energy storage systems, and lightweight structural materials, to ensure optimal integration and functionality. For instance, the choice of high-efficiency monocrystalline silicon solar cells is pivotal for maximizing energy capture from sunlight, while advanced lithium battery systems provide reliable energy storage for sustained flight operations. Additionally, the airframe materials must be lightweight yet durable to enhance aerodynamic efficiency and overall flight endurance. By carefully considering these factors, engineers can create UAVs that not only harness solar energy effectively but also meet the rigorous demands of diverse applications such as environmental monitoring, surveillance, and disaster response, ultimately contributing to a more sustainable aviation future.

5.1 DC motor

Brushless DC motors (BLDC) are widely used in Quadcopter. These consist of a permanent magnet which rotates around a fixed armature and has many advantages over brushed DC motors like more torque, reduced noise, high reliability, longer battery life and increased efficiency for less power and low weight.

- Motor KV: 1000
- Current Capacity: 12A/ 60S
- Li-PO Batteries: 2S-3S
- Max. Efficiency Current: 4-10A
- Shaft Diameter: 3.17 mm



Figure 5.1: . Brushless DC motors (BLDC)(1000kv) [12]

5.2 Propellers

Propeller is a type of fan that converts rotational motion into thrust. Generally, propellers are classified on the basis of their diameter and pitch and are represented in terms of product of diameter and pitch. For e.g. 10 *4.7, 10*4.5, etc. The diameter of propeller indicates the virtual circle that the prop generates whereas the pitch indicates the amount of travel per single rotation of propeller.

- Propeller taken in this drone is 8inch 4.5 pitch SF Propellers Black

Specification:

- Length: 8"
- Pitch: 4.5"
- Weight: 14 gm
- Shaft Diameter: 5 mm.
- Weight: 18g



Figure 5.2: Propellers [12]

5.3 Electronic Speed Controllers (ESC's)

Voltage and current supplied by microcontroller are not enough to drive the Quadcopter. A motor driver is required to supply required amount of voltage and current and this process is done by electronic speed controller (ESC). Four 30A ESCs are used in the proposed Quadcopter to control the voltage applied to the BLDC motor as per the power management signals received from the microcontroller digital pins. As 30A ESC is used, the maximum current it can handle is 23A.

- Model: SIMONK 30A.
- Constant Current: 30A (Max 40A < 10 sec).
- BEC: 5V 2A.



Figure 5.3: ESC [12]

5.4 Lithium-Ion Polymer (Li Po) Battery

These are the most popular types of rechargeable batteries for portable electronics, with a high energy density, tiny memory effect and low self-discharge. It is used at a discharging rate of 11.1V (3S).

- Model No: ORANGE 2200/3S-30C
- Weight: 175.0g
- Voltage: 11.1V
- Dimensions: 23x34x106(mm)
- Max Continuous Discharge: 30C (66.0A)
- Balance Plug: JST-XH
- Max Burst Discharge: 60C (132.0A)
- Discharge Plug: XT-60
- Charge Rate: 1-3C Recommended, 5C Max



Figure 5.4: Li-Po Battery [12]

5.5 Channel Transmitter-Receiver

The 6 channel Transmitter (Tx) and Receiver (Rx) system allows the Quadcopter to be remotely controlled through a wireless signal. Quadcopter controls would typically include throttle, pitch, roll, yaw, and mode settings. A 6-channel receiver with 2.4 GHz system is used which perfectly bonds

with the 2.4 GHz transmitter for the better performance of Quadcopter since it would not experience signal conflicts from other radio frequency (RF) controllers.

- Transmitter: FS-i6X (6 Channels)
- Receiver: FS-iA10B (10 Channels)
- RF Range: 2.408-2.475 GHz
- RF Power: < 20 dBm
- RF Channel: 135
- Bandwidth: 500 kHz
- Total Weight: 392 gm
- Power: 6V DC 1.5A



Figure 5.5: RC and receiver [12]

5.6 Flight controller

To maintain balance, the quadcopter should continuously take measurements from the sensors and make adjustment accordingly to the speed of the rotors to keep the body level. Flying capabilities and cost are the two main factors to be considered while selecting flight controller.

Flying capabilities consists of following basic factors-

- **Gyro stabilization:** It is the ability to keep the copter stable and level under the pilot control.
- **Self-leveling:** It is the ability to automatically adjust itself during any orientation so that the copter stays level.
- **Altitude hold:** It is the ability to hover at a certain distance from ground without having to manually adjust the throttle.

In here the flight controller used are **Arduino UNO and MPU 6050**

5.6.1 MPU-6050 (3-Axis Accelerometer and Gyro Sensor)

- Chip built-in 16bit AD converter, 16-bit data output
- Driver Chip: MPU6050
- Operating Voltage: 3-5V DC
- Communication: I2C/IIC Protocol
- Gyro Range: $\pm 250, 500, 1000, 2000$ °/s
- Accelerometer Range: $\pm 2 \pm 4 \pm 8 \pm 16$ g

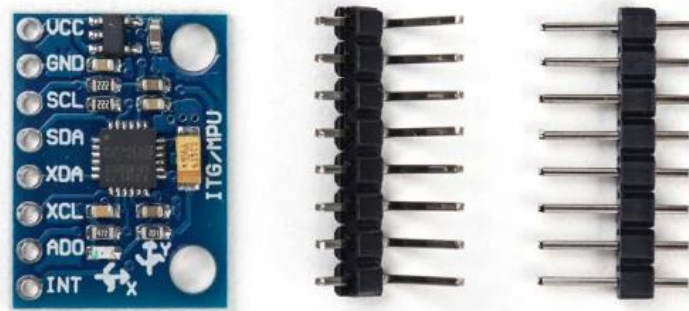


Figure 5.6: MPU 6050 [12]

5.6.2 Arduino Uno R3

- Micro-controller: ATmega328
- Operating Voltage: 5V
- Input Voltage (recommended): 7-12V
- Digital I/O Pins: 14 (of which 6 provide PWM output)
- Analog Input Pins: 6



Figure 5.7: Arduino UNO [12]

5.7 Solar cells

Solar cells are used to get more energy from sun that can increase range and endurance of the drone and here we have used 5 solar cells for recharging the battery life.

- Item weight: 10g
- Maxi voltage: 6V
- Maxi power: 100 W
- Dimension: 7*7*3 cm



Figure 5.8: Solar cell [12]

5.8 MPPT

Specification: CN3791 12V MPPT Solar Charger Module

This MPPT is used to extract the energy given by the solar cell and it provides the only required capacity of the battery so that it doesn't exceed the capacity

- Maximum Power Point Tracking (MPPT) Solar Charger Module
- 2-Pin JST Connectors for Solar Panel and Battery Connection
- PWM Switching Frequency of 300KHz
- Maximum Charging Current of 2A

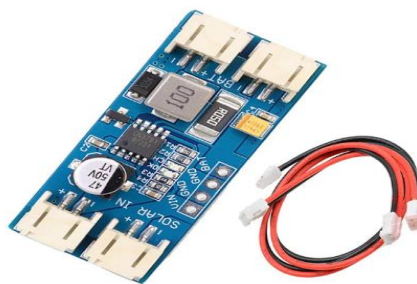


Figure 5.9: MPPT [12]

5.9 Power distribution Board

Specification: RIDEN PDB-XT60 BEC 5V&12V Power Distribution Board 3-4S 30X30mm

This board is basically for the distribution of the power to the different electronic things so that each and all components get the required power from the given battery.



Figure 5.10: PDB [12]

5.10 Body Frame

Specification: 3D PLA Printing Filament 1.75mm

This filament is used for printing the arms and body plate and solar cell holder of the drone so that it can reduce much weight of the drone and give more stability and strength to the drone.



Figure 5.11: PLA filament [12]

CHAPTER 6

DETAIL DESIGN

The detail design phase for solar-powered unmanned aerial vehicles (UAVs) is a pivotal step that encompasses the comprehensive integration of various components and systems to optimize performance and efficiency. This phase involves meticulous planning and engineering to ensure that the UAV can effectively harness solar energy while maintaining aerodynamic integrity and structural stability. Key considerations include the selection of high-efficiency photovoltaic cells, such as monocrystalline silicon, which maximize energy capture, and the strategic arrangement of these cells on the airframe to enhance power generation without compromising flight characteristics. Additionally, this stage addresses the design of energy storage systems, such as lithium batteries, which serve as buffers during low sunlight conditions, ensuring continuous operation. By employing advanced computational methods for aerodynamic analysis and structural optimization, engineers can refine the UAV's design to achieve an ideal balance between weight, power-to-weight ratios, and overall endurance. This detailed design process ultimately aims to create a robust solar-powered UAV capable of sustained flight for various applications, from environmental monitoring to surveillance missions.

6.1 Parts Design

After completing the preliminary design and conceptual design the last stage for design is detail design of the drone. In this detail design stage, all the parameters and calculation are taken which is done in above two stages of design and after that all the given and data got from above two design phases are considered and then here complete design is done.

6.1.1 Arms of the drone

Drone arm is design in such a way that the propellers doesn't hit the electronic things kept at the body of drone and its thickness and length are given according to the motor vibration analysis and strength so that it can resist the force generated during the flight. Below shown is the dimension of its length and thickness of the arm.

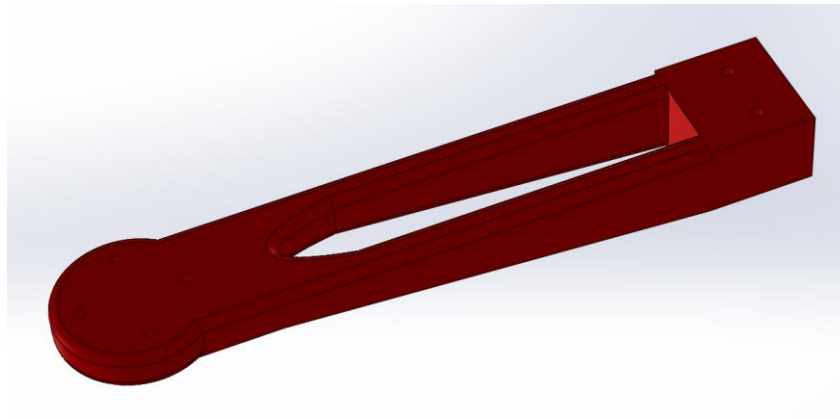


Figure 6.1: Arm

6.1.2. Top Plate

The plate is designed in such a way that it can accommodate more electronic things and made more holes to reduce the weight of the drone and connect arms and bottom plate of the drone and also the solar cell plate.

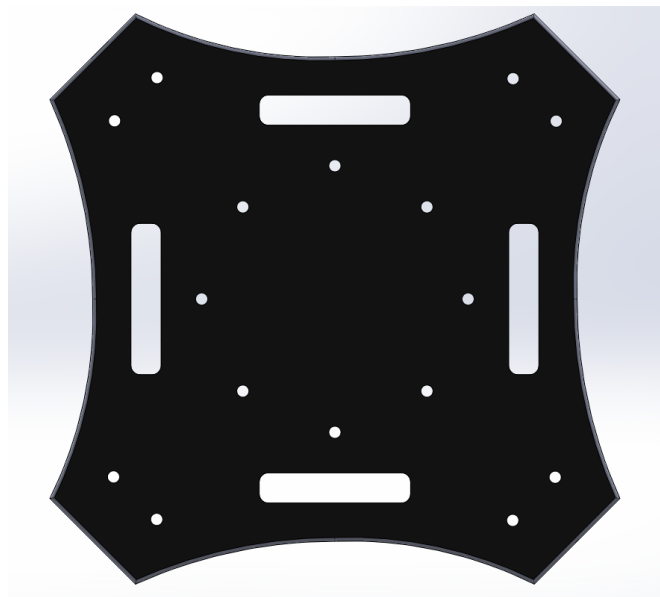


Figure 6.2: Base Top

6.1.3. Bottom Plate

This plate is design for joining the landing gear and also to cover up the bottom of drone and to strengthen the arm strength of the drone and it is different from the top plate because it is only for landing gear and battery holding and also arm connection. Also, some electronics are kept on bottom plate.

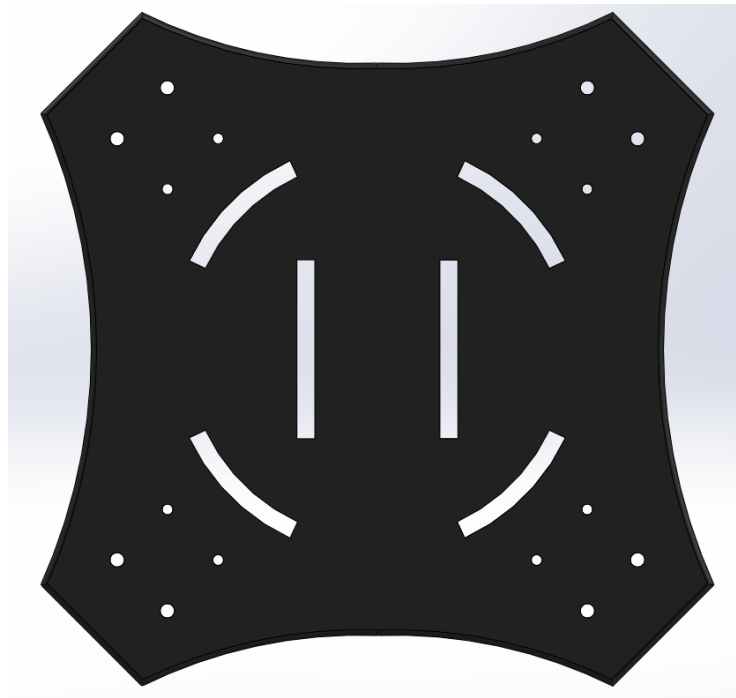


Figure 6.3: Base Bottom

6.1.4. Solar cell plate

This plate is basically for holding the solar cell above the top plate so that it doesn't get short circuit and to make sure it gets proper sun light so that it can generate the more solar energy from the sun. And it is kept at certain height to keep away from hitting by the propeller during the flight.

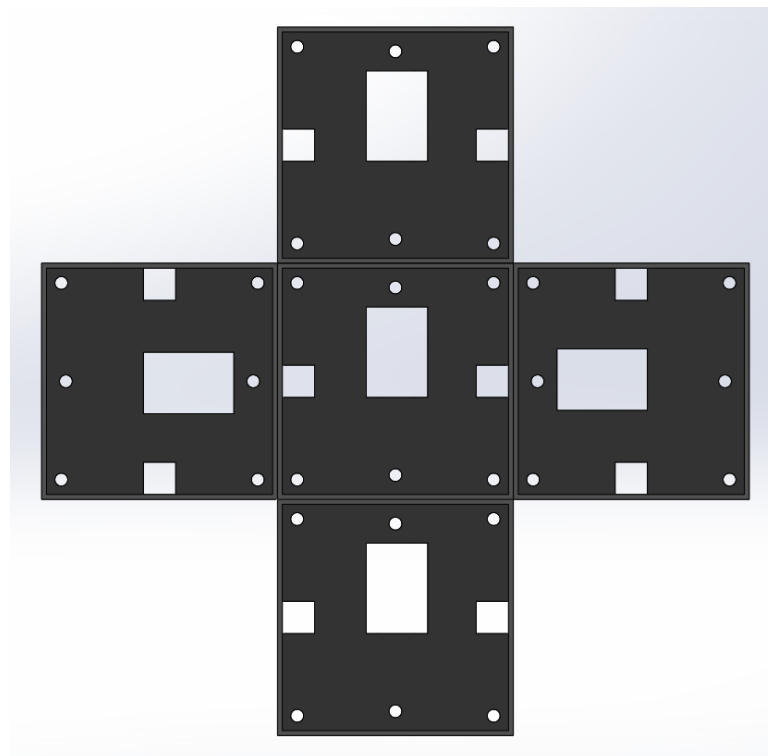


Figure 6.4: Solar Cell Plate

6.2 Parts Assembly

6.2.1 Procedure

1. **Arm and bottom plate:** All arm and plate are joined by using the M3 40mm nut and bolt screw but before doing that all the necessary circuit connections are done and the placed at the center of the plate so that its center of gravity and lies at the same point.
2. **Arm and top plate:** These two parts are connected with M3 40mm screw only and Arduino and MPU6050 is kept on it. With the use of double tap, it is pasted on the top plate.
3. **Top plate and solar cell plate:** This two is also used the same screw but in between we use some pipes to hold up the solar cell plate so that it doesn't shifts and breaks.
4. On the solar cell plate all the solar cell are pasted and connected in series connection and then it's connected to MPPT for storage.
5. **Bottom plate and landing gear:** This two are connected with M3 screw with 20mm nut and bolt so that it doesn't get too much weight on the landing gear and plate.

6.2.2 Final Design After Assembly



Figure 6.5: Final Assembly

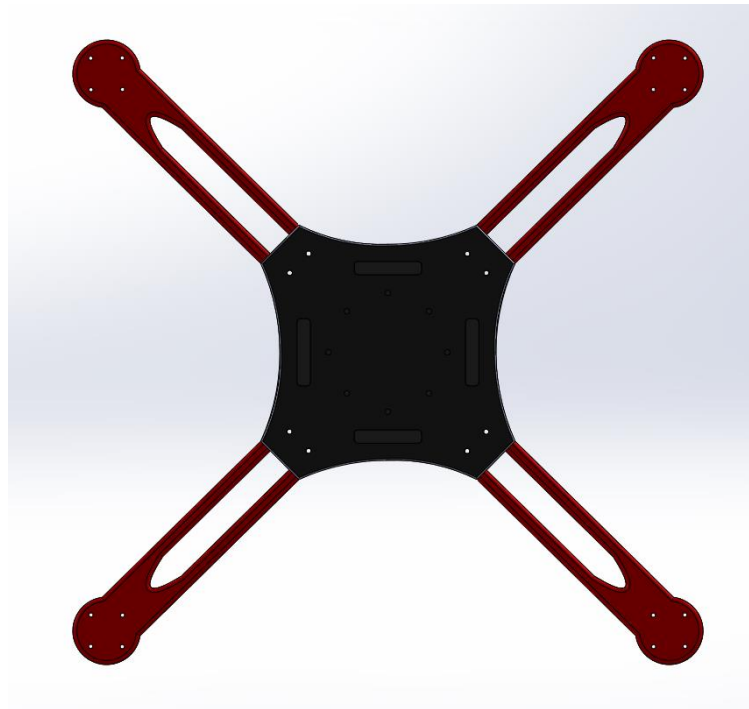


Figure 6.6: Assembly of Arm and Plate only

6.3 Structural Analysis

This report presents the findings of a structural analysis performed on drone frame. The objective of this analysis was to assess the structure's response to static loading, including stress distribution, strain, and displacement. The analysis seeks to identify high-stress concentration areas and to verify that the structure can withstand expected load conditions, maintaining its integrity over prolonged usage.

The frame that has been designed must now be validated for its strength and rigidity under the load of 1N in each arm of drone. The design model has been imported into ANSYS workbench from SolidWorks where the static structural analysis has been performed. The analysis was performed for the plastic, PLA material for fabrication of the drone.

We constructed a three-dimensional model of the drone using CAD software, defining its geometry, materials, and other physical properties. The model included the frame, supports, and other critical components of the drone. We then applied loads to the model to simulate and analysed the model's response to these loads, identifying areas of high stress and potential failure. Based on these results, we optimized the design of the drone, using ANSYS to simulate and evaluate different configurations and materials.

1. Equivalent Elastic Strain

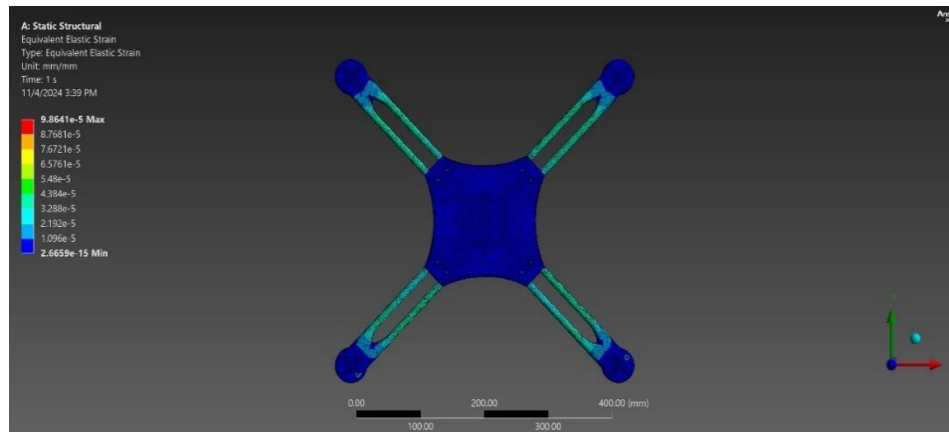


Figure 6.7: Equivalent Elastic Strain

Equivalent Elastic Strain, often represented as ϵ_{eqv} , is a measure of strain that combines the deformation effects in all directions into a single scalar value. It's calculated using the von Mises strain criterion, which provides an equivalent strain as if it were uniaxial, making it useful for evaluating complex stress states. Here, on applying 1N load force on each arm of the drone obtained maximum equivalent elastic strain is 9.8641×10^{-5} and minimum eqv. Elastic strain is 2.6659×10^{-15} . This result helps in assessing the overall deformation behavior of the structure. High equivalent strain values can indicate regions experiencing significant distortion, which might be areas of concern in terms of material durability or failure.

2. Maximum Principal Elastic Strain

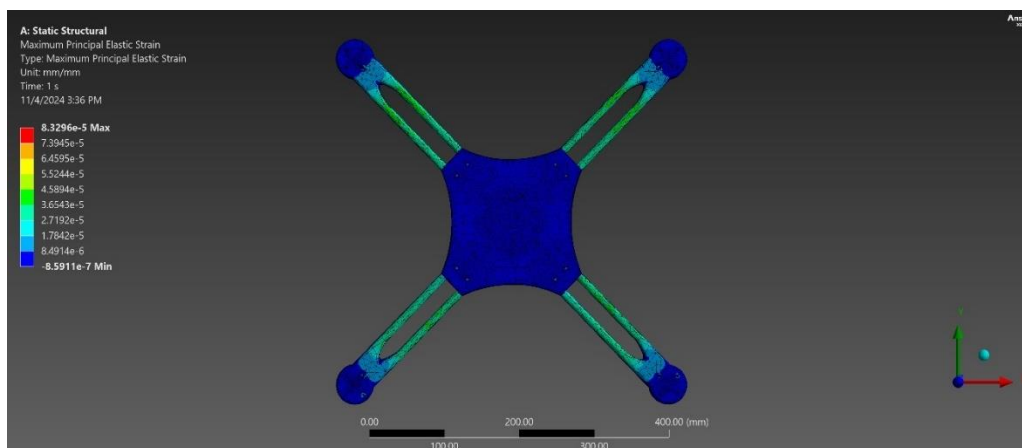


Figure 6.8: Maximum Principal Elastic Strain

The Maximum Principal Elastic Strain refers to the maximum strain along the principal axis where the material is most stretched or compressed. Principal strains are directions in which the normal strain (deformation per unit length) reaches its peak. Here, maximum principal elastic strain is 8.3296×10^{-5} . This measure gives insight into the strain that could lead to material failure, especially in

brittle materials where tensile strength limits may be critical. Regions with high maximum principal elastic strain indicate where the material is experiencing the greatest extension or compression. These areas could be prone to cracking or failure if the strain exceeds material limits.

3. Maximum Principal Stress

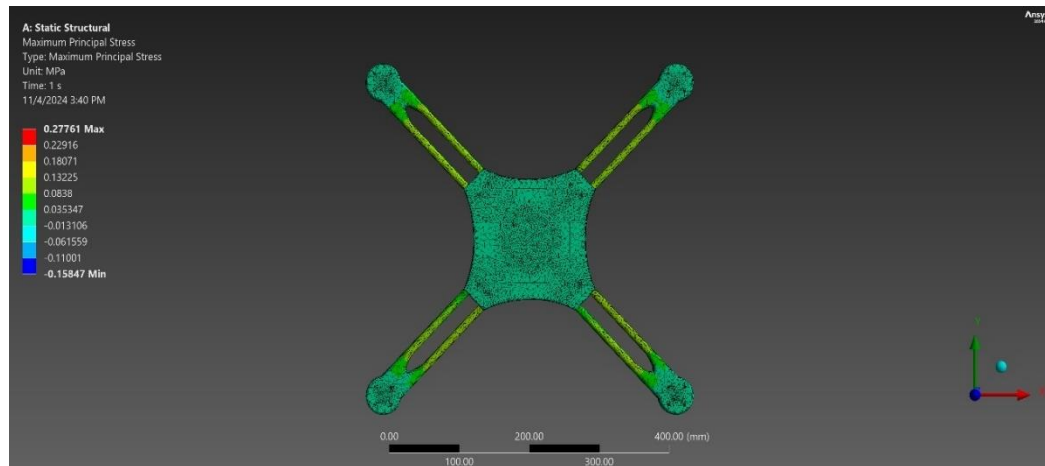


Figure 6.9: Maximum Principal Stress

Maximum Principal Stress is the highest normal stress acting on the structure in a specific principal direction. It is one of the most critical stresses to analyze in materials prone to brittle failure, as they often fail along the direction of maximum principal stress. Here, max principal stress is found to be 0.27761. The maximum principal stress is useful in determining whether a material might fracture or fail, particularly in cases where the material has different strengths in tension and compression. High values in this result indicate areas at risk for failure under tensile loading. In your seaplane structure, zones with elevated maximum principal stress might need reinforcement to avoid cracks or fractures.

4. Strain Energy

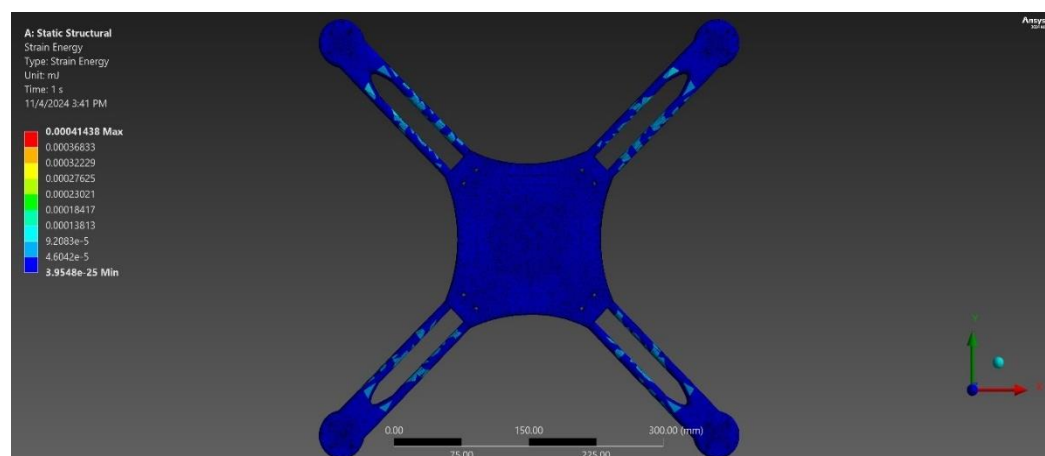


Figure 6.10: Strain Energy

Strain Energy refers to the energy stored in the material as it deforms under load. This is calculated based on the stress and strain in the material and represents the energy absorbed by the structure up to that point. Here strain energy is 0.00041438J. Strain energy density can be a useful metric to identify regions where a lot of energy is being absorbed and, therefore, may indicate potential locations of failure initiation. Areas with high strain energy are those where the material is storing significant deformation energy, which could be an indication of stress concentration points. These areas should be examined for possible design adjustments or material strengthening if they are near or above critical limits.

5. Total Deformation

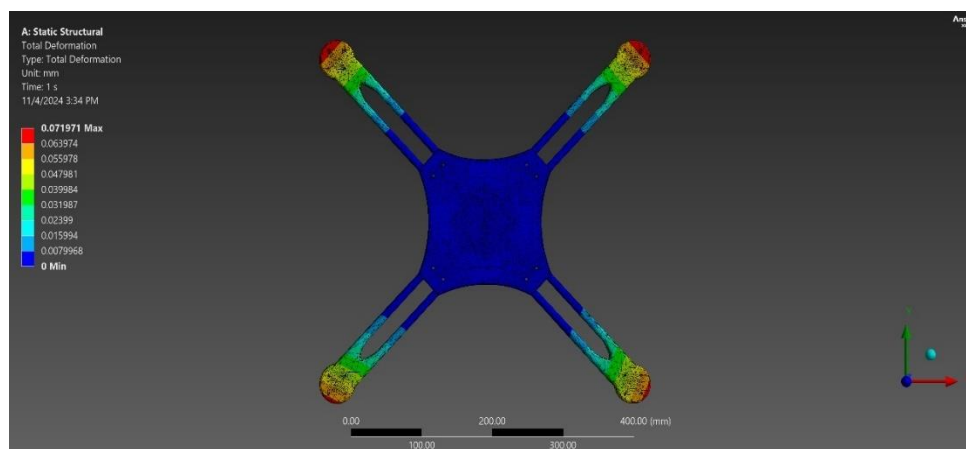


Figure 6.11: Total Deformation

The maximum deformation of 0.071971 mm indicates the largest displacement of any point in the drone frame under the applied load. The minimum deformation of 0 mm suggests that some areas, likely at fixed boundary or support locations, experienced no displacement, serving as stable anchor points. A deformation of 0.071971 mm is relatively small, indicating that the frame has a high rigidity and structural integrity under the current load. This suggests the drone frame can withstand operational loads with minimal deflection, contributing to stability during flight. Since deformation is limited, the likelihood of failure due to excessive bending or buckling is low. This is a favourable result for the frame's durability and reliability.

Since deformation is limited, the likelihood of failure due to excessive bending or buckling is low. This is a favorable result for the frame's durability and reliability. If additional loads were anticipated (e.g., from higher thrust or additional components), the frame could likely handle them without substantial modifications. The total deformation analysis indicates that the drone frame is stable and well-suited for flight, with minimal deflection under load. This contributes positively to the frame's operational robustness and efficiency.

CHAPTER 7

DEVELOPMENT AND ASSEMBLY

The development and assembly of solar-powered unmanned aerial vehicles (UAVs) represent a critical phase in transforming theoretical designs into functional prototypes capable of efficient flight. This stage involves the meticulous integration of various components, including high-efficiency photovoltaic cells, lightweight structural materials, and advanced energy storage systems. Engineers must ensure that each element is optimized for performance and reliability, addressing challenges such as weight distribution, aerodynamics, and energy management. The assembly process typically includes the installation of solar panels on the airframe, which are strategically positioned to maximize sunlight exposure while maintaining aerodynamic efficiency. Additionally, the incorporation of systems like Maximum Power Point Tracking (MPPT) allows for real-time adjustments to optimize energy capture and utilization during flight. Through rigorous testing and iterative refinement, the development phase culminates in a UAV that not only harnesses solar energy effectively but also meets the operational demands of various applications, paving the way for sustainable aviation solutions.

7.1 Components Assembly

Components fixed on the Arm:

All the motors are placed at the tip of the arm and ESC are placed at the centre of the arm as shown below picture.

Components fixed at top plate:

Arduino and power distribution board is attached in this top plate where all the connections are done and the receiver is kept at this plate only.

Components fixed at bottom plate:

Battery is placed below the bottom plate and landing gear also attached here so that it supports all the weight of the drone.

7.2 Circuit Description and Assembly

7.2.1 Circuit Connection and Its Description

1. Solar cells to battery

The solar cells are firstly connected to the MPPT (solar cell connector) then from other sides of MPPT (battery connector) the battery is connected so that the solar cells power or energy that is converted and regulated at the given capacity of the battery (ensuring to provide right amount of the energy to the battery).

2. Battery to power distribution board

The battery provides the power to the power distribution board where this board distributes all the power required by the ESC and other electronics. This board plays very important by distributing the required voltage to all the ESC, motor and flight controller. Battery voltage is divided by the power distribution board.

3. Power distribution board to ESC

From this board only the ESC gets its required power and then this ESC regulate the power required by the DC motor so that it doesn't gets out of its limit and burnt it. ESC directly receives the required power from PDB then it again regulates itself to provide required power to motor.

4. Flight controller set up using Arduino UNO and MPU6050

- I. **Arduino and MPU 6050 connection:** Arduino SDC and VCL is connected to the SDC and VCL of MPU6050 and also the GNC and VCC of MPU6050 is connected to the GNC and 5V of the Arduino.
- II. **Arduino and ESC signal connector:** All the ESC signal connector are connected to the Arduino board on no. 4, 5, 6, and 7 and ESC is also connected to the ground of the Arduino.
- III. **Arduino and receiver connection:** 5V from receiver is connected to the Arduino and 5V from Arduino is given to the MPU6050. Using jumper wire channel 1, 2, 3, and 4 is connected to the hole no. 8, 9, 10, and 11 on Arduino board.

5. Arduino UNO coding

After done with all the connection on Arduino board then the coding is done for calibration and regulation of the board so that it doesn't misbehave when given command from the RC and to have more smooth flying of the drone i.e. yawing, rolling and many movements of it.

Wire used for connection

Wire used for the connection of the Arduino board and other electronics is jumper wire that is for female and male jumper wires. Soldering is done for each connection to make sure it can give proper current passage and smooth flow of the current.

7.2.2 Complete circuit diagram

Below shown diagram is the complete drone circuit diagram and all the connections are well soldered and well connected.

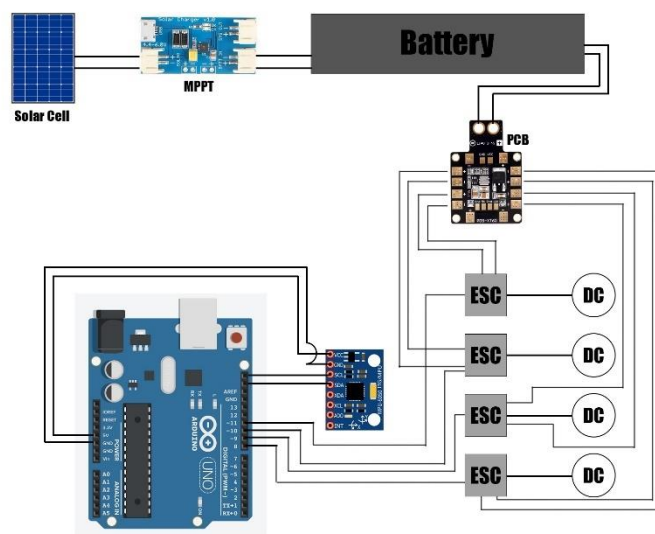


Figure 7.1: Complete Circuit Diagram

CHAPTER 8

RESULT AND CONCLUSION

8.1 Result

Though, successfully demonstrated the feasibility of a solar-powered UAV that can operate autonomously with enhanced endurance. The integration of solar technology not only reduces reliance on conventional fuels but also contributes to sustainable aviation practices. Future work may focus on optimizing the design further and exploring advanced materials and technologies to improve efficiency and performance in varying operational scenarios. This project highlights the potential of renewable energy sources in transforming UAV technology, paving the way for more sustainable aerial applications in fields such as surveillance, environmental monitoring, and logistics.

The integration of a solar power system significantly impacts the overall flight duration of UAVs (Unmanned Aerial Vehicles) by enhancing energy efficiency and extending operational time.



Figure 8.1: Phase-1 of Assembling 3-D printed body frame



Figure 8.2: Implemented Solar Panels over the top view of the UAV body frame

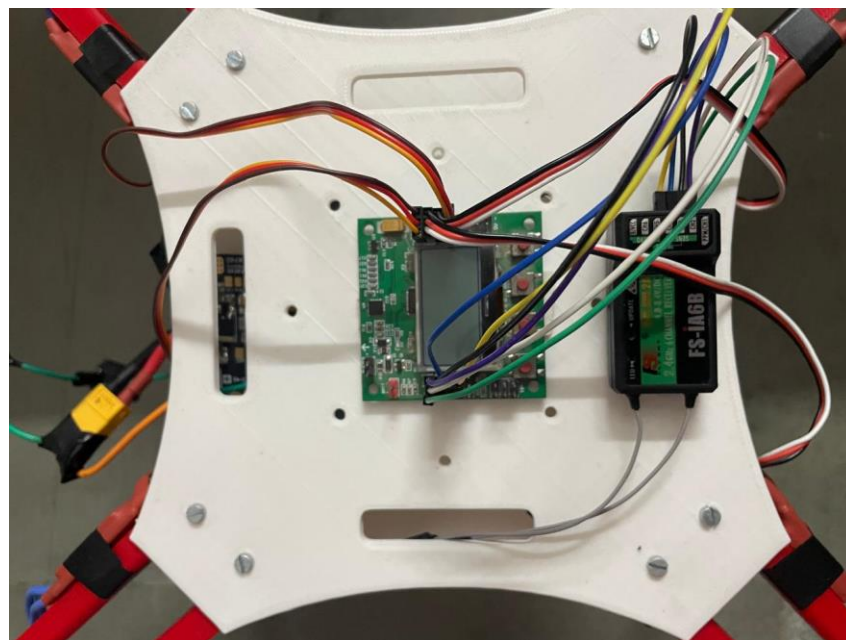


Figure 8.3: Assembly of electrical components inculcated with wire connections



Figure 8.4: Attached Propellers along with solar panels, DC Motors



Figure 8.5: Top View of Solar Powered UAV

8.2 Conclusion

The development and implementation of a solar power system in UAVs represent a transformative approach to aerial technology, addressing the critical challenges of energy consumption and environmental sustainability. Throughout this project, we successfully designed a solar-powered UAV that integrates photovoltaic cells to harness solar energy, effectively increasing flight duration and reducing reliance on conventional battery systems. Result indicates that the solar power system can save an average of 22.5% of battery capacity during flight, allowing for longer missions without the need for frequent recharges. This capability is crucial for applications requiring extended aerial operations, such as environmental monitoring, surveillance, and disaster management. The optimization of energy management techniques, including the use of Maximum Power Point Tracking (MPPT), further enhances the efficiency of energy collection and utilization. Moreover, our experiments confirmed that solar-powered UAVs can operate autonomously for over 24 hours, significantly surpassing the capabilities of traditional fuel-based or battery-only UAVs. This not only underscores the feasibility of solar energy in aviation but also highlights its potential to reduce carbon emissions and promote sustainable practices in various industries. In conclusion, the project successfully meets its objectives by demonstrating that solar-powered UAVs are not only viable but also advantageous for future aerial applications. Continued research and development in this area will likely lead to further innovations, improving efficiency, payload capacity, and overall performance. The findings from this project pave the way for a new era of sustainable aviation technology that leverages renewable energy sources to meet the growing demands for longer and more efficient flight capabilities.

8.3 Future Scope

Future advancements in solar-powered UAV technology are set to significantly enhance their capabilities and applications across various sectors. Key developments include improved solar panel efficiency through innovations like ultra-thin perovskite solar cells, which promise higher power outputs while reducing weight. The integration of artificial intelligence (AI) will optimize flight paths and energy management, allowing UAVs to adapt dynamically to changing weather conditions for maximum solar exposure. Advanced energy management systems utilizing machine learning will enhance resource allocation, ensuring continuous operation during missions. Additionally, hybrid power systems that combine solar energy with batteries or fuel cells will provide greater flexibility and reliability. As technology progresses, the applications of solar-powered UAVs will broaden, encompassing areas such as agriculture for crop monitoring, environmental monitoring for wildlife tracking and disaster assessment, and infrastructure inspection for surveying hard-to-reach areas. Enhanced communication systems will enable real-time data transmission over long distances, improving mission coordination. Overall, these advancements will not only extend flight durations but also make solar UAVs invaluable tools in promoting sustainability and efficiency in aerial operations across diverse fields. Expected Advancements

- **Improved Solar Panel Efficiency:** Ongoing research aims to develop more efficient solar panels that can generate higher power outputs even in low-light conditions. Innovations such as ultra-thin perovskite solar cells, which are lightweight and flexible, promise to increase energy absorption while minimizing weight, thus enhancing overall flight performance and endurance.
- **Integration of Artificial Intelligence (AI):** The incorporation of AI algorithms is expected to optimize flight paths and energy management. AI can analyze real-time data to adjust the UAV's trajectory for maximum solar exposure, improving efficiency and extending flight durations. This capability allows drones to autonomously adapt to changing weather conditions and optimize battery usage dynamically.
- **Advanced Energy Management Systems:** Future UAVs will likely feature sophisticated energy management systems that utilize machine learning for intelligent resource allocation. These systems will enhance the UAV's ability to harvest solar energy effectively, manage battery life, and ensure continuous operation during missions.

8.4 Challenges

The integrating solar power into UAVs presents several significant challenges that must be addressed to enhance their feasibility and performance. One of the primary challenges is weather dependency, as solar-powered UAVs rely heavily on sunlight for energy generation. Adverse weather conditions, such as cloud cover or rain, can drastically reduce their operational efficiency and flight duration. Additionally, the requirement for large surface areas to accommodate solar panels poses a design constraint; larger panels can capture more energy but also increase the drone's weight and size, complicating aerodynamics and manoeuvrability. Another critical challenge is achieving high efficiency in solar energy conversion. Current solar panels are not 100% efficient and optimizing their placement on the UAV to maximize exposure to sunlight while maintaining structural integrity is complex. The need for lightweight materials that do not compromise the drone's stability further complicates the design process. Furthermore, initial costs associated with high-quality solar panels and advanced energy management systems can be prohibitive, particularly for smaller companies or projects with limited budgets. Maintenance is another concern; solar panels require regular cleaning to ensure optimal performance, which can be difficult in remote locations. Additionally, there are regulatory hurdles that UAV operators must navigate, including compliance with aviation laws and privacy concerns when flying over residential areas. Overall, addressing these challenges involves a multifaceted approach that combines advancements in technology, innovative design strategies, and careful consideration of operational environments to fully realize the potential of solar-powered UAVs.

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APPENDIX

COST ANALYSIS

Table 3: Cost Analysis

S.NO.	DESCRIPTION OF GOODS	UNITS	AMOUNT(₹)
1.	ESC 30A BLDC (Simonk)	4	1,560
2.	Quadcopter Landing Gear	1	190
3.	GY521/MPU6050 Gyroscope Accelerometer	1	120
4.	PLA Red Filament (1.75 kg)	1	900
5.	Jumper Cable 1M-1F	10	20
6.	Jumper Cable 1M-1M	10	20
7.	Berg/Header Male	10	50
8.	Double Sided Tape Roll (1m)	1	15
9.	3 x 10 Machine Screw Packet	1	30
10.	Power Distribution Board with XT60	1	290
11.	Solar Panel 04V 100mw 70x70	5	300
12.	Propeller 8045 8 in	2	180
13.	Power Distribution Board with XT60	1	290
14.	FlySky FS-16 2.4G 6Ch Remote Control	1	4,750
15.	Jumper Cable 1F-1F	10	20
16.	A2212 1000 KV BLDC for Drone (Soldered Connector)	4	1,400
17.	Orange 11.1 V 2200 mAh 30C 3S Lithium Polymer Battery Pack	1	1,524
18.	CN3791 12V MPPT Solar Charger Module	1	260

TOTAL

11,811

SPECIFICATIONS

$$\begin{aligned}
 1. \quad W_e &= W_{st} + W_{pp} + W_{sys} \\
 &= 380 + 288 + 407 \\
 &= 1075 \text{gm} / 1.075 \text{Kg}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad &\text{Take-off gross weight,} \\
 W_o &= W_e + W_{bat} + W_{pay} \\
 &= 1075 + 175 + 250 \\
 &= 1500 \text{gm} / 1.5 \text{Kg}
 \end{aligned}$$

$$\begin{aligned}
 3. \quad &\text{Empty Weight fraction,} \\
 W_e / W_o &= 1.075 / 1.5 \\
 &= 0.716
 \end{aligned}$$

4. Solar cell: output of 20V
5. Motor: 12A/60S
6. ESC: 30A
7. Battery: 2200mAh, 24.1V
8. RC range: 1500m